

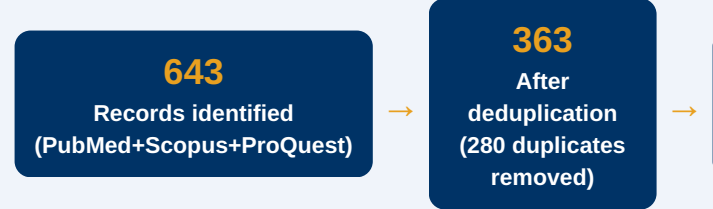
EFFICACY AND SAFETY OF PUSH-DOSE NOREPINEPHRINE FOR  
INTRAOPERATIVE HYPOTENSION MANAGEMENT IN NON-OBSTETRIC  
SURGERY: A SYSTEMATIC REVIEW*Rsu Senoaji* | Department of Anesthesiology, [Institution Name]

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## METHODS

- Design:** Systematic review (PRISMA 2020)
- Databases:** PubMed, Scopus, ProQuest
- Keywords:** "Norepinephrine" AND ("push dose" OR "bolus") AND ("intraoperative hypotension" OR "perioperative" OR "surgery")
- Inclusion:** Adults ≥18 yrs, non-obstetric surgery, push-dose/bolus NE for IOH, RCT design
- Exclusion:** ICU/sepsis, obstetric, single case reports, reviews
- Quality:** RoB 2 (Cochrane)

## PRISMA FLOW



RESULTS – SAFETY

| Parameter                                      | NE Bolus                                             | Comparator   | p        |
|------------------------------------------------|------------------------------------------------------|--------------|----------|
| Overshoot hypertension AUC >100 (Thomsen 2026) | 42.0 (18–87) mmHg·min                                | 47.4 (9–131) | 0.390 NS |
| Overshoot hypertension AUC >100 (Vokuhl 2025)  | 16 (6–53) mmHg·min                                   | 20 (2–72)    | 0.986 NS |
| Arrhythmia (all studies)                       | Not reported as structured outcome in any study      |              |          |
| Vascular access complications                  | None documented; peripheral IV safe in all 3 studies |              |          |

RISK OF BIAS (ROB 2)

| Study        | D1 | D2 | D3 |
|--------------|----|----|----|
| Hassani 2018 |    |    |    |
| Thomsen 2026 |    |    |    |
| Vokuhl 2025  |    |    |    |

D1=Randomization · D2=Deviations · D3=Missing data · D4=Outcome mea  
Concerns inherent to open-label design; no study rated high risk.



REFERENCES (MAX 7)

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6. Page MJ, et al. The PRISMA 2020 statement. *BMJ.* 2021.

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